

Industrial Measurement Data Platform

Case Study

(By Julian Cabrera Marceglia)

Background

A large automotive manufacturing operation relied on dimensional inspection data generated by multiple measurement systems.

As production complexity increased, the existing reporting process became difficult to maintain and scale. Inspection data was distributed across multiple files and reporting activities required significant manual effort.

A new manufacturing program introduced approximately five times more part-configuration combinations, creating the need for a standardized and scalable solution.

Challenge

The organization needed to:

- Centralize measurement data from multiple sources.
- Standardize reporting workflows.
- Improve traceability.
- Support SPC and capability studies.
- Scale to significantly higher product complexity.
- Reduce manual reporting effort.

Solution

I independently designed and implemented a centralized measurement-data platform composed of *three applications*:

Loader

Responsible for:

- Parsing measurement results from multiple systems.
- Validating data.
- Calculating measurement accuracy.
- Determining pass/reject status.
- Storing results in a centralized repository.

Reporter

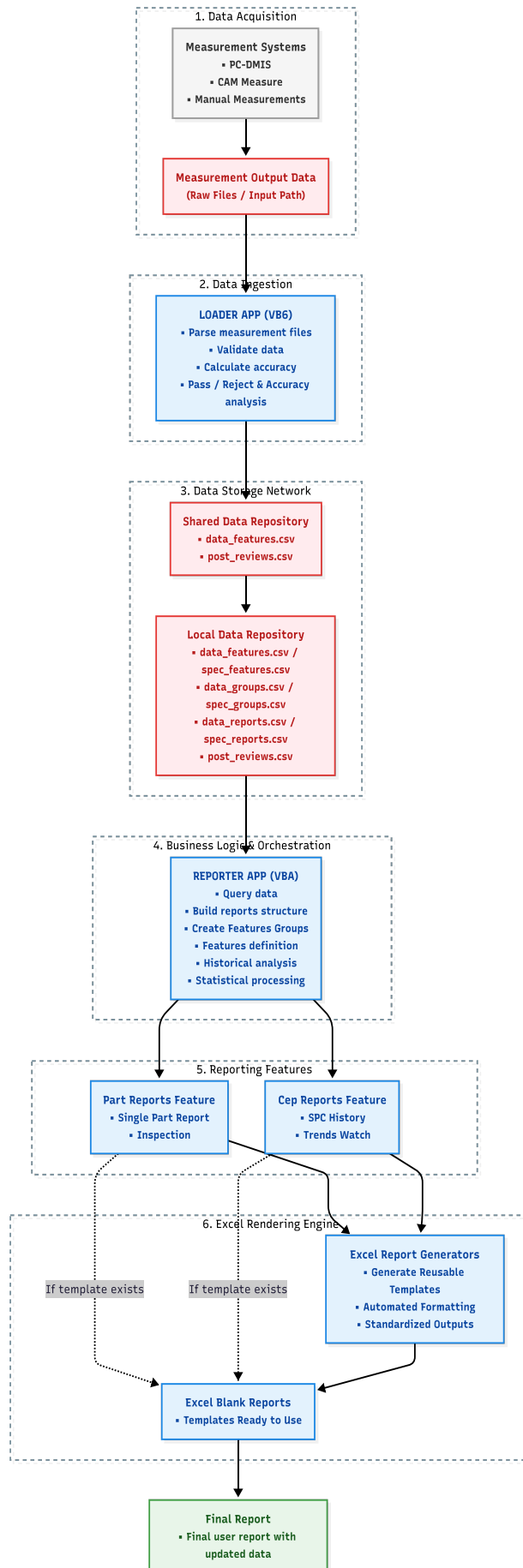
Responsible for:

- Generating individual measurement reports.
- Generating historical statistical reports.
- Querying centralized measurement data.
- Delivering standardized outputs.

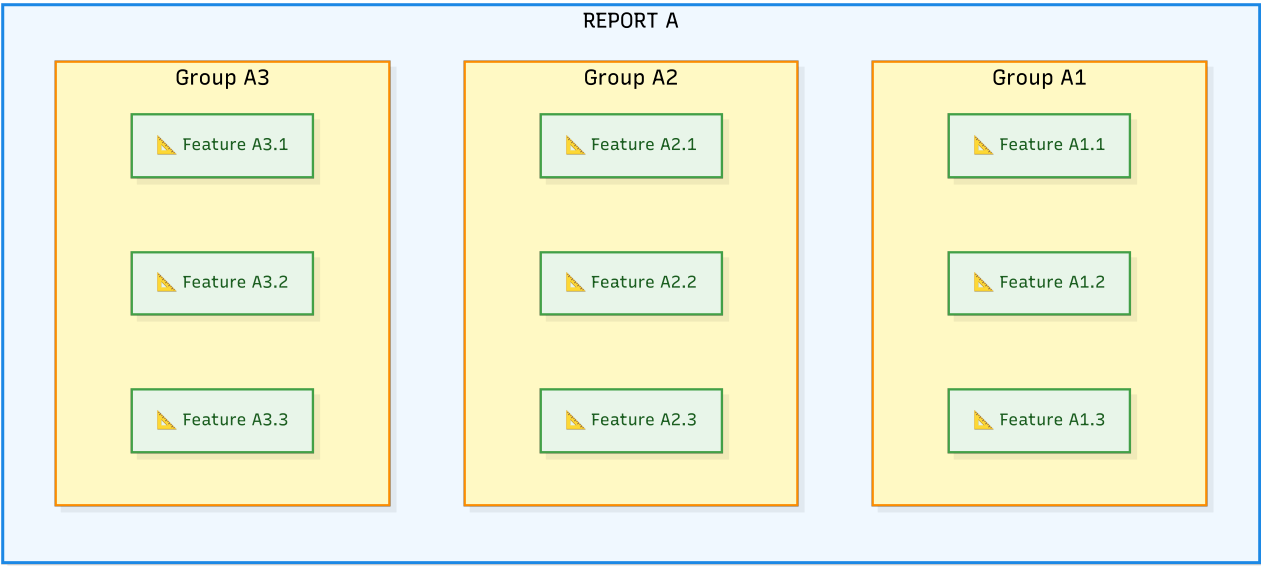
Report Generators

- Reusable Excel-based report templates capable of automatically generating standardized reports from centralized data.

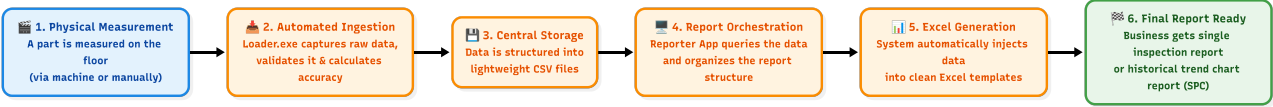
General Architecture:



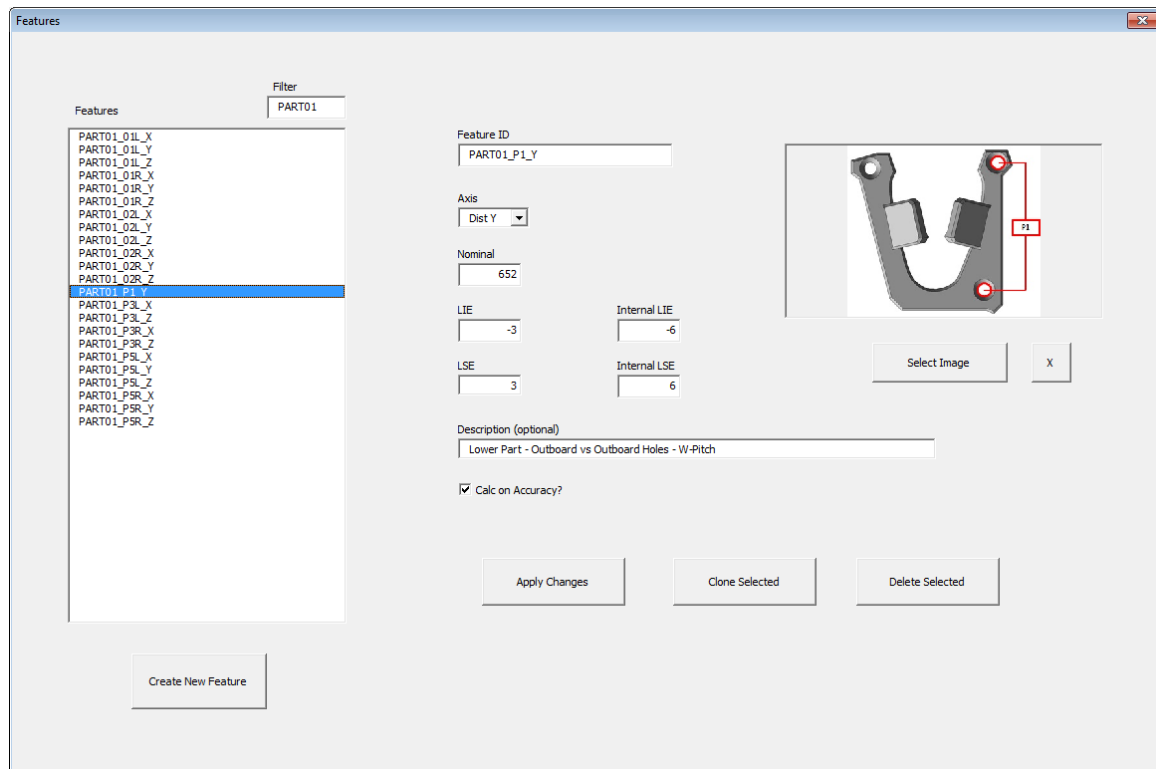
Data Model:



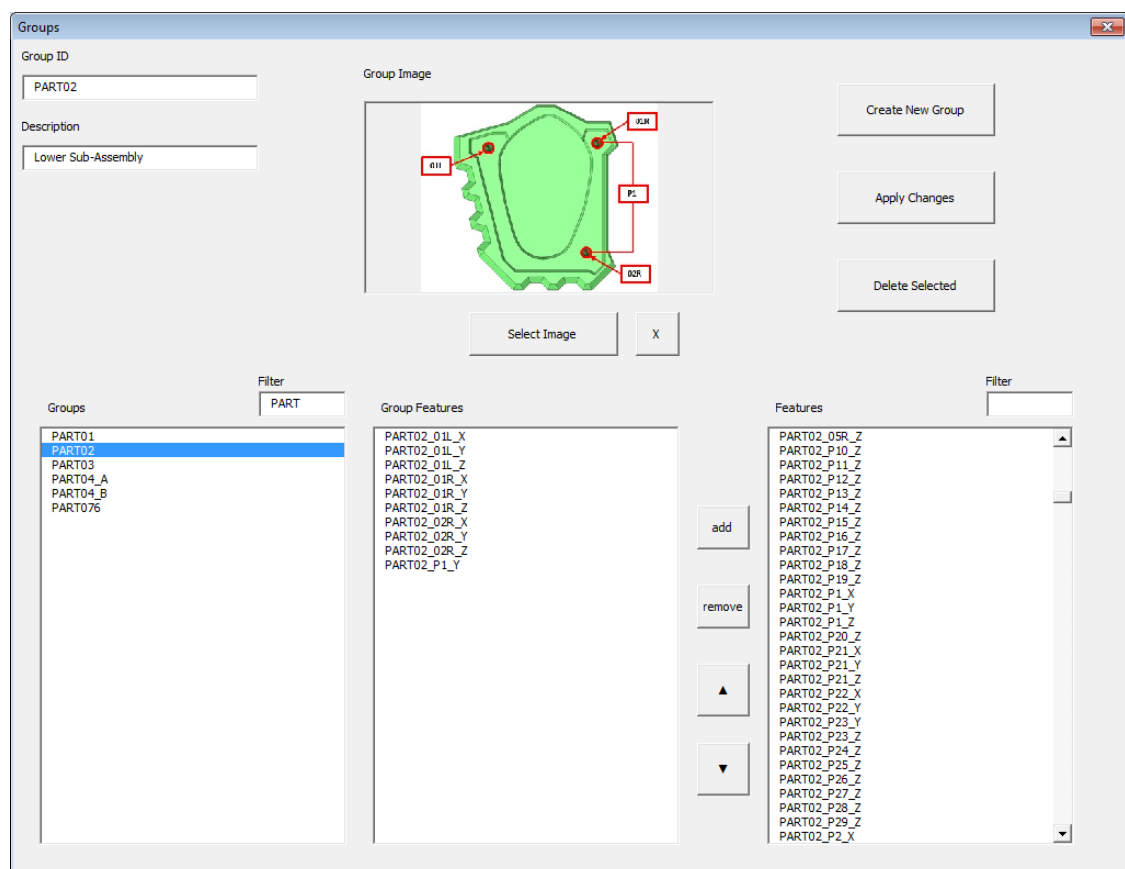
Operational Flow



Screenshots



features menu



groups menu

Report ID

REP_PART

Type

Part report

Description

Part Model N - Part Report

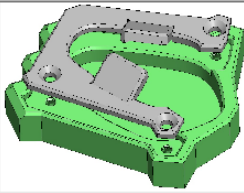
Device Id

DEA_TORO

Part Id

X234-44

Report Image



Select Image

X

Create New Report

Apply Changes

Delete Selected

Reports

Filter

PART_CEP_REP

REP_PART

Report Groups

PART01

PART02

add

remove

▲

▼

Groups

Filter

PART01

PART02

PART03

PART04_A

PART04_B

PART076

reports menu

Measures

Refresh Data

Date	Time	Part Id	Part num	To Cep	Device Id	Accuracy	Status	Part Name
13/06/2026	11:26:39	X234-44	812873	False	DEA_TORO	96.1%	Rejected	Part Model N
13/06/2026	09:18:00	X234-44	823003	True	DEA_TORO	95.5%	Rejected	Part Model N
12/06/2026	20:16:49	X234-44	797183	False	DEA_TORO	97.4%	Pass	Part Model N
12/06/2026	17:48:32	X234-44	822777	True	DEA_TORO	95.8%	Rejected	Part Model N
12/06/2026	15:31:51	X234-44	822737	True	DEA_TORO	96.1%	Pass	Part Model N
12/06/2026	10:03:44	X234-44	798733	False	DEA_TORO	90.8%	Rejected	Part Model N
12/06/2026	10:03:44	X234-44	798733	False	DEA_TORO	92.7%	Pass	Part Model N
12/06/2026	07:44:43	X234-44	822558	True	DEA_TORO	96.1%	Pass	Part Model N
11/06/2026	19:43:36	X234-44	821923	False	DEA_TORO	96.9%	Pass	Part Model N
11/06/2026	17:46:10	X234-44	822352	True	DEA_TORO	96.1%	Pass	Part Model N
11/06/2026	15:49:46	X234-44	822306	True	DEA_TORO	95.3%	Rejected	Part Model N
11/06/2026	07:39:04	X234-44	822143	True	DEA_TORO	96.6%	Pass	Part Model N
10/06/2026	19:51:27	X234-44	791959	False	DEA_TORO	97.6%	Rejected	Part Model N
10/06/2026	17:42:16	X234-44	821931	True	DEA_TORO	94.8%	Pass	Part Model N

X234-44

Apply Filters

Available Reports

REP_PART

Description: Part Model N - Part Report

Generate Part Report

part report generation menu

Generate Cep Report

Available Cep Reports

PART_CEP_REP

Description: Part Model N - Cep Report

By N Measures

60

By Date Range

Generate Cep Report

Description: Part Model N - Cep Report

By N Measures

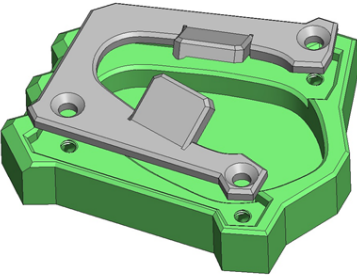
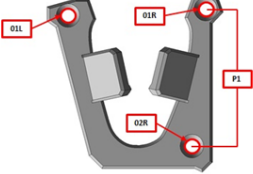
By Date Range

From10/08/2025domingo, 10 de agosto de 2025

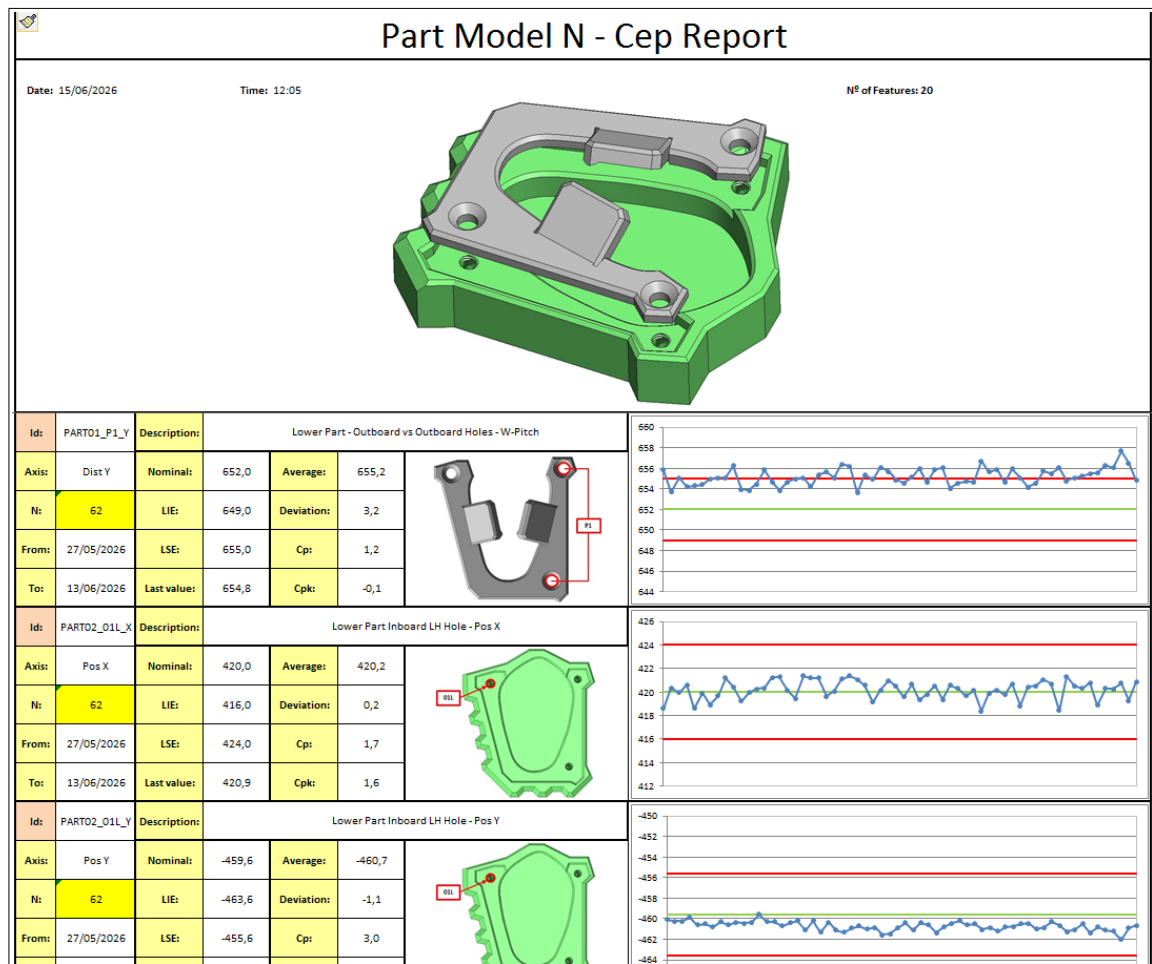
To11/05/2026lunes, 11 de mayo de 2026

Generate Cep Report

cep report generation menu (by N and date range)

Part Model N - Part Report																							
Date: 13/06/2026			Time: 11:26:39			Part Id: X234-44			Part Number: 812873														
																							
accuracy 100,0%			condition 																				
Upper Sub-Assembly																							
																							
Feature Id	Axis	Value	Nominal	LIE	LSE	Deviation	Judge	OOT	Description														
PART01_01L_X	Pos X	226,9	226,2	-3,0	3,0	0,7			Upper Part Inboard LH Hole - Pos X														
PART01_01L_Y	Pos Y	-328,2	-326,0	-3,0	3,0	-2,2			Upper Part Inboard LH Hole - Pos Y														
PART01_01L_Z	Pos Z	989,9	990,2	-3,0	3,0	-0,3			Upper Part Inboard LH Hole - Pos Z														
PART01_01R_X	Pos X	226,4	226,2	-3,0	3,0	0,2			Upper Part Inboard RH Hole - Pos X														
PART01_01R_Y	Pos Y	326,7	326,0	-3,0	3,0	0,7			Upper Part Inboard RH Hole - Pos Y														
PART01_01R_Z	Pos Z	989,2	990,2	-3,0	3,0	-1			Upper Part Inboard RH Hole - Pos Z														
PART01_02R_X	Pos X	228,9	226,2	-3,0	3,0	2,7			Upper Part Outboard RH Hole - Pos X														
PART01_02R_Y	Pos Y	449,1	449,0	-3,0	3,0	0,1			Upper Part Outboard RH Hole - Pos Y														
PART01_02R_Z	Pos Z	1051,5	1052,3	-3,0	3,0	-0,8			Upper Part Outboard RH Hole - Pos Z														
PART01_P1_Y	Dist Y	655,0	652,0	-3,0	3,0	3			Lower Part - Outboard vs Outboard Holes - W-Pitch														
Lower Sub-Assembly																							
																							

part report example



cep report example

Full reports examples: <https://drive.google.com/file/d/1lXTQSSh6w0LrvR36xYBI7Uf2-IPXQNRa/view?usp=sharing>

Results

- Centralized more than 50,000 measurement records.
- Supported multiple dimensional measurement systems.
- Reduced report generation effort by approximately 10x.
- Improved standardization and traceability.
- Enabled scalable support for future manufacturing programs.
- Adopted by quality and manufacturing personnel in production operations.

Technologies

- Visual Basic 6
- VBA
- SQL Queries
- Excel Automation
- SPC
- Cp/Cpk Analysis
- GD&T
- CMM Data Processing
- Industrial Reporting Systems